

Pricing an ABS Challenge

Decision Thresholds Under Uncertainty in Major League Baseball

Introduction

Major League Baseball's Automated Ball-Strike challenge system gives a player only seconds to decide whether an umpire missed a call. A successful challenge corrects the call and preserves the team's challenge inventory. An unsuccessful challenge costs one of a limited number of challenge units. Passing preserves that resource for later but may allow an incorrect call to stand. This research asks whether that decision can be valued when it is made rather than judged by what happens afterward.

Methods

We introduce **Challenge Threshold**, a break-even framework built from expected runs. Each pitch occurs within a game state defined by factors including count, outs, baserunners, inning, score state, and remaining challenge inventory. We estimate the expected-run value of the state if the call stands and if it is corrected. Their difference provides the immediate correction value, V . We separately estimate the future expected-run value, C , of the challenge unit lost following an unsuccessful challenge. The break-even probability is $C/(V + C)$: the probability of an incorrect call at which challenging and preserving the challenge have equal modeled expected value.

We implement the framework through a reproducible reference specification, **Challenge Threshold Standard 1 (CT-S1)**, and introduce **Decision Opportunity Score (DOS)**, an inverted 0-100 representation in which higher scores indicate more favorable opportunities to act. DOS is not a probability or measure of player confidence.

CT-S1 was developed using MLB games from March 25 through September 9, 2026, frozen, and evaluated on an untouched September 11-27 confirmation period.

Results

Among 22,584 supported game states, median absolute movement in Challenge Threshold between development and confirmation was 0.76 percentage points; the 90th percentile was 2.61 points, with rank correlation of 0.995. The reference therefore reproduced closely when its definition remained fixed.

We also tested eight prespecified modeling conventions. Among 22,428 states supported under all eight, approximately 98% had a spread greater than 10 percentage points between the lowest and highest thresholds. In one representative state, CT-S1 estimated 67.3%, with a 64.9-69.7% conditional sampling interval, while alternative conventions ranged from 58.7% to 75.7%.

Conclusion

ABS separates two components often confused when decisions are judged afterward: the value of the opportunity and the strength of the evidence available to the decision-maker. Challenge Threshold prices the former without observing the latter. More broadly, ABS provides a measurable implementation of a Decision Threshold: the point at which evidence must become strong enough for the expected value of acting to exceed the value of preserving the alternative.